Initiative Vision: verified driving scenarios from a real car

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A camera and the CAN bus of a real car record a drive. An agent on Nebius Token Factory reviews the moments that simple rules flag, and confirms, corrects or rejects each one, citing frames and CAN samples that code then checks. The events that the agent and a human labeller agree on become MATLAB driving scenarios. This report explains how it works, how it was evaluated, and where it fails. Every number comes from the repository (github.com/osfieldgaga/initiative-vision-nebius) and can be reproduced from it without an API key.

> **In one paragraph**
> The existing OpenCV pipeline (lanes, lane offset, stop signs) runs over a recorded drive and is joined with the car's CAN signals on a common clock. Rules propose 37 *candidate* events. For each, NVIDIA Nemotron 3 Super on Token Factory holds a short tool-using conversation (reading the data, asking Nemotron 3.5 Super VL about frames, re-running OpenCV at full frame rate) and emits a verdict with *claims*: citations of the form "in frame 4592 the speed was 0.0 mph". A deterministic validator checks every claim against the source and recomputes the key facts from the raw CAN log; the model can fix refused answers, and what never passes is marked unverified. Every model exchange is recorded and replayable. Against 15 hand-labelled events, the agent raises precision from 0.32 (rules) to 0.36-0.48 over 4 runs, at a cost in recall (0.60-0.67). The 8 events that agent and labels agree on are exported as `drivingScenario` files, and all of them, plus the whole drive, replay the CAN speed within 0.05 m/s rms in MATLAB.

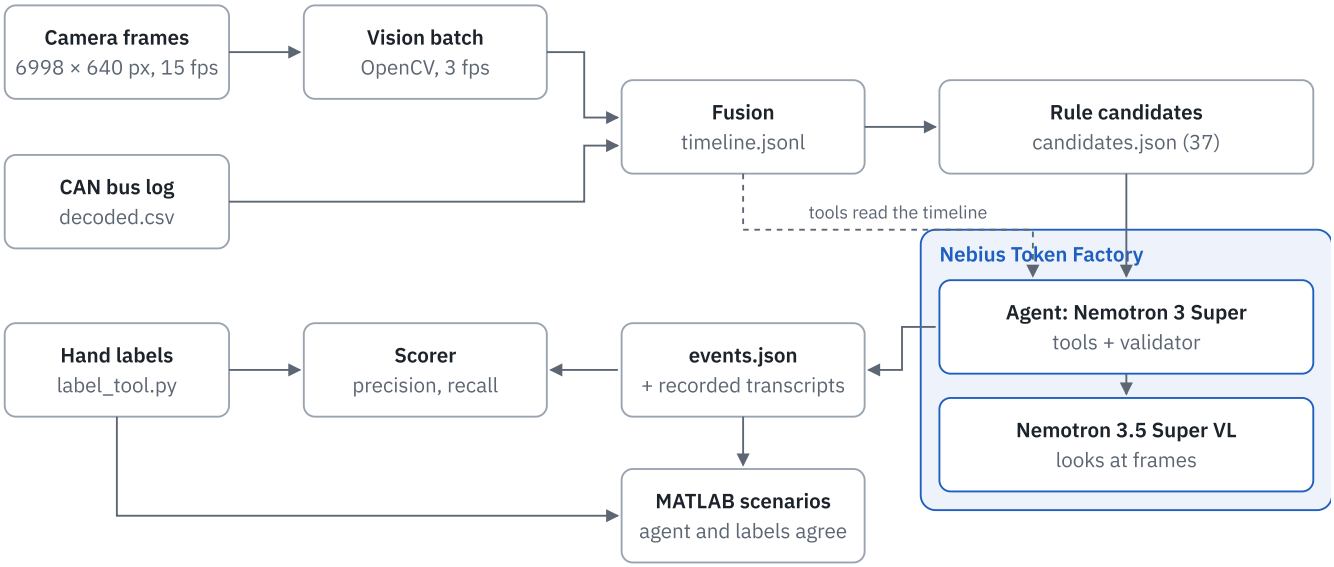
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1. Goal and constraints

Engineers who test driver-assistance software need lists of real moments ("rolled through the stop sign", "left turn without a blinker") to turn into test cases. Today people scrub through video to find them. This project does it with an agent, under four constraints set at the start:

- **Traceable and evaluated.** Every agent output must be traceable to the source frames and CAN samples, with an evaluation showing how often it is correct. This drives the validator, the claims, the recordings and the hand labels.
- **Runnable by judges** without the car or any cloud account: everything works from the shared privacy copy of one drive, and the agent replays offline from its recordings.
- **Don't break the OpenCV entry.** The project builds on my OpenCV AI Competition 2026 entry, which is being judged separately. All work is in a new repository; the vision code (`lld.py`, `signs.py`) is imported unchanged.
- **Bounded cost:** the account started with $1 of credit; every run has a spend cap.

2. Architecture



Only the agent calls a model; everything else is deterministic code that writes a file. The scorer compares the rules' candidates and the agent's events with the same hand labels; the MATLAB export takes only the events both agree on.

- **Rules find, the model judges.** Cheap rules scan the whole drive and are tuned to miss nothing. The model only looks at the 37 short moments they flag, which keeps a full run at about $0.17 and gives it one focused question at a time.
- **The model never computes facts that code can compute.** Speeds, minimums, blinker timing and steering direction come from the data. The model's job is judgement (is this a parking-lot manoeuvre? is that red shape a stop sign?), and it must back each judgement with citations that code verifies.
- **Every stage writes a file** that can be inspected, tested and reused: `vision.jsonl`, `timeline.jsonl`, `candidates.json`, `events.json`, the `.m` scenarios.

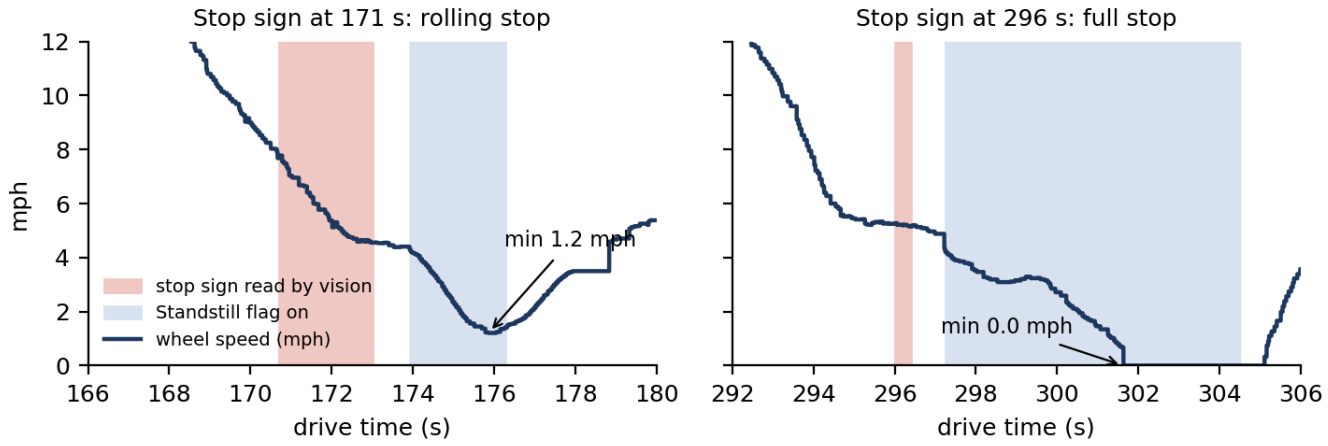
3. The data

One drive (Oct 22, 2026) recorded by a Raspberry Pi 3B in my car: 6998 frames at 640 × 640 (15 fps, 461 s) and the CAN log. Judges get its **privacy copy**: faces and number plates blurred, the GPS columns removed,

and the raw CAN file (which holds the VIN) left out. All times are **drive time**, seconds since the first frame; frames and CAN rows share the Pi's clock. The camera is mounted sideways; frames are rotated upright and undistorted by `lld.undistort_frame` before anyone, person or model, looks at them.

CAN signal names don't come with trustworthy units, so each signal used was checked on the drive itself:

Signal	Meaning, as verified	How
`EBCMWheelSpdFront`	speed, m/s	ratio to `VehicleSpeed` is 2.237 = mph per m/s
`SteeringWheelAngle`	degrees, positive = left	every blinker episode is followed by steering the same way
`TurnSignals`	1 = left, 2 = right	same check
`Standstill`	**not** "speed is zero"	see below
`YawRate`	(raw − 2.5) × 0.112 deg/s, positive = left	calibrated against the original drive's GPS heading (167 windows, r = 0.96); used only for the MATLAB paths



Wheel speed at the two stop signs. At 171 s the car's Standstill flag (blue) comes on at 4.3 mph and the speed never reaches zero (slowest 1.2 mph); at 296 s the speed is 0.0 for about 3 s. Stop details are therefore computed from wheel speed, never from the flag.

4. From a drive to candidates

- **Vision batch** (`vision_batch.py`) runs `lld.process_frame` (YOLOPv2 on `cv.dnn`, bird's-eye lane geometry, the stop-sign reader) at 3 fps: 1511 frames. Check: on the 95 lane frames labelled for the OpenCV entry, `lane_valid` matches its published evaluation exactly (precision 0.977, recall 0.729). The sign reader fires at both real stop signs and once on red autumn leaves (seq 3856), a false positive deliberately left in as a test of the agent.
- **Fusion** (`fuse.py`) attaches to each frame the car's state at that instant: the last CAN row at or before it, never a future value. Each row keeps `row_ts_ns`, the timestamp of the exact CAN row used: the hook for traceability.
- **Candidates** (`candidates.py`) apply simple rules: a standstill gives a stop (a stop sign if one was read in the 10 s before); steering past 90° gives a turn, 20-90° for 2 s at speed a curve; a vision departure flag in 2 consecutive frames a lane departure. 37 candidates: 19 turns, 7 stops, 6 lane departures, 3 stop signs, 2 curves. Each carries its evidence (frames, key frames, CAN rows, statistics).

5. The agent

`agent.py` holds **one short conversation per candidate** with Nemotron 3 Super at temperature 0. That keeps each context small and cheap, makes every decision independent and individually replayable, and lets a failure on one candidate leave the others untouched. The system prompt gives the task (confirm, correct or reject), the meaning of each data column, the event definitions read at run time from `data/eval/LABELLING.md` (the same text the human labeller used), and rules of thumb: reject parking-lot manoeuvring; vision can be wrong, so check frames; a stop sign the car didn't stop at is corrected to `no_stop`, not rejected.

Tool	What it does	Limits
`get_window(t_start, t_end)`	timeline rows as a compact table: frame, time, CAN row, mph, steering, blinker, standstill, brake, gear, accel, lane_valid, offset, departure, signs	≤ 60 s, ≤ 60 rows
`look_at_frames(seqs, question)`	asks Nemotron 3.5 Super VL about upright frames; it answers one line per frame	≤ 4 frames
`rerun_vision(t_start, t_end)`	runs the OpenCV pipeline on every frame (15 fps) of a short window	≤ 3 s, cached
`emit_event(...)`	the answer: verdict, type, times, side, detail, 1-6 claims, a reason without numbers	validated

A **claim** is a citation the validator can check: `{"seq": 4592, "field": "mph", "value": "0.0"}`. Fields are the table's columns plus `vision_stop_sign` (did OpenCV read a stop sign in that frame) and `frame_check` (a quote of what the VL model said about that frame). The model calls tools for up to 8 turns. When it calls `emit_event` the validator checks the answer; refused answers go back with the list of errors, and after 4 tries the candidate is kept as the rules found it and marked **unverified**. The system never invents an answer of its own.

6. The validator

Language models are good at judgement and unreliable with precise facts: they misread a table, round numbers, or state something plausible that isn't in the data. `validate()` is plain code that turns "the model says so" into "the data says so":

1. **Form:** allowed verdict, type and detail; times inside the drive; 1-6 claims.
2. **Every claim against the source:** the frame must exist; the value must equal the data for that frame (within 0.6 mph, 2° of steering); the frame must be near the event. A `frame_check` must quote the VL model's answer *about that frame*.
3. **Facts recomputed from the full-rate CAN log**, with the model's own start and end: the stop detail from the slowest wheel speed (0 → full, under 5 mph → rolling); the blinker detail from whether it was on in the 8 s before the start; the side from the steering actually reaching 45° (turn) or 10° (curve) that way.
4. **Vision gate:** a stop sign or lane departure can only be accepted after the agent has looked at frames of that moment.

> **What this guarantees, and what it doesn't**
> Guaranteed: every number and citation in an accepted event matches the source files; stop, blinker and side details agree with the CAN log; no vision-based event is accepted without a visual check. Not guaranteed: that the model's *judgement* is right, nor that its conclusion follows from what it saw (section 11 shows a case where it didn't). That is what the scoring measures.

In the final run the validator refused 13 answers on 11 of 37 candidates, and every one was then fixed: too many claims (6), a claimed value that contradicts the data (6: OpenCV's stop-sign reading, the brake, the gear, the standstill flag), a frame with no data (2) and an unknown field (1); one refusal can list several errors. The validator is unit-tested on the real drive in `src/tests/test_agent.py`.

7. Walkthrough: the stop sign at 296 s

Candidate `c23`, from the final run's recording `runs/…/agent/transcripts/c23.json` (5 calls). The rules proposed: stop sign read at 296.0 s, standstill until 304.4 s, detail `full_stop`.

```
1. Super  (thinking) "verify stop sign detection around 296 sec"   → get_window(295, 305)
   code   21 rows: speed 5.7 → 0.0 mph, sign "stop" read at frame 4507, left blinker on

2. Super  → look_at_frames([4507], "Is there a stop sign visible in this frame?")
   VL     "Frame 4507: Yes"

3. Super  → emit_event(confirmed, stop_sign, 296.04-304.41, full_stop, claims:
              4507 vision_stop_sign true · 4592 mph 0.0 · 4528 standstill true · 4627 standstill false)
   validator  REFUSED: "claim seq 4627: standstill is True in the data, not 'false'"

4. Super  (thinking) "we need to adjust claim: at seq 4627, standstill is true (since still stopped)"
          → emit_event(... same, with 4627 standstill true)
   validator  ok: all 4 claims match the data; vision gate passed (frame 4507 checked);
              slowest speed in 296.04-304.41 s is 0.0 mph → full_stop ✓
```

The model's first answer contained a plausible but false fact (that the car had started moving at frame 4627). The validator caught it, the model corrected it, and the accepted event in `events.json` resolves every claim to its source, so anyone can open the frame and the CAN line:

```
{"seq": 4592, "field": "mph", "value": "0.0", "t_s": 302.007,
 "frame": "00004592_4851274309000.jpg", "can_row": 4851272192169}
```

This event is one of the 8 exported to MATLAB (`ev_c23_stop_sign.m`, section 13); its header names hand label row 22 as agreeing.

8. Record and replay

`tokenfactory.py` (152 lines, standard library only) sends each request to Token Factory's OpenAI-compatible API and appends to a transcript the **SHA-256 of the exact request**, the request (images replaced by their hash) and the full response. With `--replay` it answers each request from the transcript, in order, after checking the hash.

- **Judges need no key:** the transcripts, the timeline and the cached re-runs are committed. A clone plus the drive download reproduces the agent's run, for free, in a few seconds.
- **It shows the run is genuine:** a replay only succeeds if the code, the prompts, the data and the tool results are exactly those of the recorded run; otherwise it stops with "call k differs from the recording". Replaying the final run reproduces `events.json` byte for byte.
- **One allowance:** frames sent to the VL model are JPEGs this code encodes, and another OpenCV build may encode the same frame with different bytes. If only image bytes differ and every text in the request matches, the call is replayed and counted, and the count is printed.

Why: temperature 0 makes a model more repeatable, not deterministic (section 10 measures it); recording the responses is the only way to make "re-run it" give the same answer.

9. Ground truth

An evaluation needs a reference independent of the system: what really happened, decided by a person watching the drive. `label_events.py prefill` listed 36 moments worth a look, mostly from CAN, each with a guessed type; I reviewed them in `label_tool.py` (the OpenCV entry's virtual-Pi player plus a panel that edits the CSV), fixed types, times and details, marked non-events `no`, and added events the list missed.

- **Definitions first:** six event types with written definitions in `LABELLING.md`; the agent reads the same text. Only driving on roads counts: manoeuvring inside a parking lot or driveway is `no`.
- **A consistency fix before scoring:** my first labels marked 9 parking-lot moves and a U-turn on a short dead-end street (4 rows) as events, contradicting the definitions. I changed them to `no` before the final scoring, each with a note saying why.
- **Final labels:** 38 rows, **15 real events** (9 turns, 3 stop signs, 2 lane departures, 1 other stop), 1 of them added by hand (a slight right turn at 74 s the list had missed).
- **One disagreement kept:** I labelled the 171 s stop sign a full stop; the CAN log says the car never went below 1.2 mph (a rolling stop). At that speed a car looks stopped on video. The label stands as my judgement, and the scores show it as the one detail error.

> **Caveat: anchoring**
> The label list was pre-filled by the same CAN rules the candidates use, so the rules' recall is inflated by construction (they cannot miss what they suggested). The most trustworthy numbers are precision, detail accuracy, and recall on the hand-added events.

10. Results

A detection matches a labelled event of the **same type and side** whose time overlaps it (padded by 1 s), one to one. **Precision** = right ÷ flagged: when it reports an event, how often is it real. **Recall** = found ÷ 15: how many real events it finds. Events the agent rejected are not flagged.

Discipline: the agent fixes for the final run were chosen before labelling, and the final run was named in advance; the repeat runs measure noise and are not used to pick a better score.

	flagged	right	precision	found / 15	recall	unverified
Rules alone	37	12	0.32	12	0.80	–
Agent, run 1 (code before the fixes)	23	11	0.48	11	0.73	3
Agent, final run	21	9	0.43	9	0.60	0
Agent, repeat 3	20	9	0.45	9	0.60	0
Agent, repeat 4	25	9	0.36	9	0.60	0
Agent, repeat 5	21	10	0.48	10	0.67	0

Over the 4 runs with the final code, precision ranges 0.36-0.48 (median 0.44) and recall 0.60-0.67 (median 0.60), always above the rules' 0.32 precision and below their 0.80 recall. 20 of 37 candidates got the same decision (verdict, type and side) in every run; the other 17 split, e.g. `c06` turn right ×2 / rejected ×2; `c15`

stop_sign ×2 / rejected ×2; `c32` turn right ×3 / rejected ×1. The splits include real events (the turns c06 and c32) and the red leaves (c15, reported as a stop sign in half the runs), so no single run is representative; a majority vote over runs is an obvious next step, to be measured on a new drive.

Per type, final run

Type	labelled	rules: right / flagged	agent: right / flagged	notes
turn	9	8 / 19	6 / 12	agent false alarms are mostly turns inside parking lots or the dead-end street
stop_sign	3	2 / 3	2 / 3	the third "stop sign" is the red leaves (section 11); the 83 s sign is never in view
other_stop	1	1 / 7	1 / 3	
lane_departure	2	1 / 6	0 / 2	the one it confirmed at 372 s is on the opposite side to the label
curve	0	0 / 2	0 / 1	a real turn downgraded to a curve
all	**15**	**12 / 37**	**9 / 21**	

Detail accuracy (on matched events: full or rolling stop, blinker or not) is 0.89 for the agent and 0.92 for the rules; the only miss in the final run is the 171 s stop sign (rolling by the CAN log, full by my label). The one hand-added event (a slight right turn at 74 s) was not proposed by the rules, so neither method could find it.

> **Reading the numbers**
> The agent does what a filter should: it removes about half of the rules' false alarms, and every event it keeps carries validated evidence. It also removes some real events, so recall drops. With 15 events, one event moves recall by 7 points; the run-to-run spread is as large as the difference between the runs, so these numbers are indicative, not a benchmark.

11. Where it fails



Upright frames as the VL model sees them. **Left**, 83 s: I labelled a stop sign here, but it is never in the camera's view (driving into the sun, the sign outside the frame); vision cannot find it and the agent reports "a stop, no stop sign visible", which is true to the camera. **Middle**, frame 6039 (395 s, `c32`): a real right turn into a parking-lot entrance. Asked "is there an intersection ahead?", the VL model said no, and the agent rejected the turn, although the definitions count road ↔ parking entrance as a turn. **Right**, frame 3856 (251 s, `c15`): red leaves that OpenCV read as a stop sign.

1. **It trusts a "no" from the VL model when it shouldn't.** Mid-turn, the camera faces the new road or a driveway, not an intersection. For the real turns at 179 s and 395 s the VL model answered "no intersection", and the agent downgraded one to a curve and rejected the other. This is the same weakness measured on lane lines (section 12): the VL model is a reliable confirmer and a weak denier.
2. **It ignored a "no" it should have heeded.** For the red leaves (`c15`) the VL model answered "No" for all 6 frames it was shown, yet the agent reported a stop sign the car didn't stop at. The prompt says a sign the car didn't stop at is corrected, not rejected, and the agent applied that to a sign that wasn't there. The validator only checks *that* it looked, not that what it saw supports the answer. In run 1 the same candidate was correctly rejected.
3. **Side of a lane departure.** At 372 s the agent confirmed a departure on the right, the side the vision flag said; I labelled it as moving into the left turn lane. It counts as a false alarm and a miss.
4. **Limits of the camera.** The 83 s stop sign is invisible to the camera; no vision method could have found it.

Not fixed on purpose: fixing 1 and 2 now (a validator rule that a stop sign needs a VL "yes", a prompt line about mid-turn views) would be tuning on the same labels that score it. The honest way is to fix them on this drive and measure on a new one.

12. How far can the VL model be trusted?

The agent uses the VL model as a second pair of eyes, so it was measured. `eval_vl_lanes.py` asks Nemotron 3.5 Super VL, for each of the 95 frames hand-labelled for the OpenCV entry, whether painted lane lines are visible on both sides, and compares with the labels and with OpenCV's `lane_valid`:

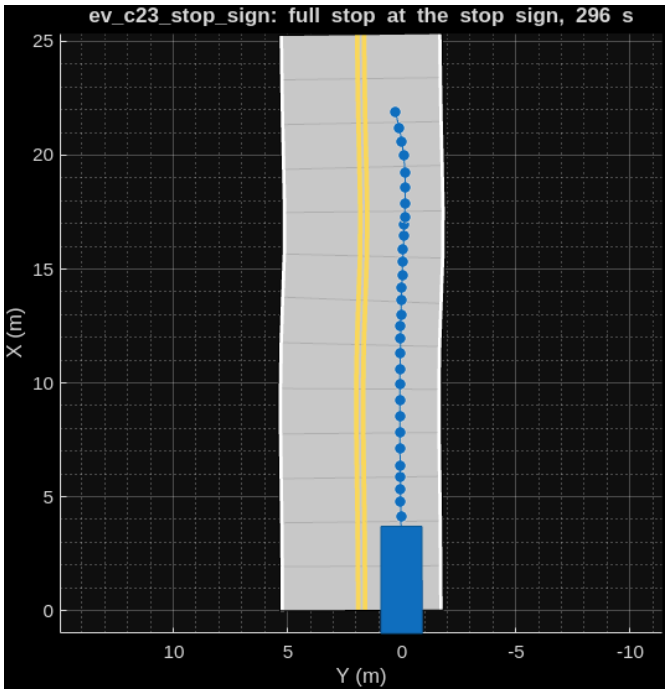
"Both lane lines visible"	precision	recall	accuracy	TP / FP / FN / TN
Nemotron 3.5 Super VL	1.00	0.25	0.54	15 / 0 / 44 / 36
OpenCV pipeline (YOLOPv2 + geometry)	0.98	0.73	0.82	43 / 1 / 16 / 35

When the VL model says lines are visible it is right; it misses three quarters of the lines that are there. The agent's prompt states these numbers, so a VL "no lines" is treated as weak evidence. Sending raw sideways frames gave recall 0.00; sending what the labeller saw (undistorted, upright) gave 0.25, so every VL call uses that view.

13. MATLAB scenarios

`export_matlab.py --labels` exports only the events that **the agent verified and the hand labels agree with** (same type, side and detail, overlapping time, exactly as the scorer matches them): fewer scenarios, each confirmed twice. That gives 8 files (6 turns, a full stop at a stop sign, a stop at the lot exit) plus `full_drive.m`, the whole drive in one scenario. Each file is a MATLAB function returning `[scenario, egoVehicle]`; run on its own it plays itself, with the drive time, speed and current event in the title.

- **Speed:** the front wheel speed, every 0.1 s; each standstill becomes one waypoint with a wait time.
- **Path:** heading = integral of the calibrated yaw rate; position = integral of speed along it, from the origin facing +x. Held while stopped, so sensor noise doesn't spin a parked car. Waypoints every 0.5 m or 1 s.
- **Road:** two 3.5 m lanes along the path, ego in the right lane. Real road geometry would need a map and the GPS, which the shared drive doesn't contain.
- **Header:** the drive, window, agent verdict and reason, every validated claim with its frame and CAN row, and the agreeing label row.





Plotted by MATLAB R2025a. Left: the stop-sign scenario of section 7 (blue dots are waypoints). Right: the whole 459 s drive, 2.2 km, as dead-reckoned from CAN; it is right locally, but the heading drifts over minutes, so it is not a map.

Checked in MATLAB. `verify_matlab.py` simulates every file with `advance()` and compares the ego's speed and position with the CAN profile it came from, with tolerances set before the first run (speed 0.5 m/s rms, timing 1 s, end position 2 m). All 9 pass: speed within 0.01-0.05 m/s rms, timing within 0.1 s, end within 0.8 m; the whole drive is 0.05 m/s rms and 0.1 s late after 459 s.

Getting there took four corrections, none of which loosened a tolerance: a checker bug (it counted the time after the last waypoint as speed zero); denser waypoints (1 m spacing cut tight turns); no fixed stop time (a car a few tenths late was cut off before its last waypoint); and **retimed stops**. The wheel-speed sensor drops from

about 0.3 m/s straight to 0, while MATLAB changes speed evenly between waypoints, so every slow-down and start took 0.3-0.6 s longer than the car did, and the whole drive fell 2.4 s behind over 8 stops. Each stop's wait is now set so the car reaches the next waypoint on time.

How close is the path to reality? Replaying the profile proves the file is faithful to the CAN data, not that the integrated path is faithful to the road. Against the original drive's GPS heading change (9 events of the first export where GPS heading is usable): median error 11.5°, worst 30°. Good for "a left turn of about 90°", not for lane-level geometry.

14. Cost

- **Meter and cap:** every response's token usage is priced (Super $0.30 / $0.90 per million input / output tokens; the VL price isn't published, so it is assumed high on purpose, $1 / $3), and the client refuses to send past `--max-usd`.
- **Compact contexts:** one conversation per candidate, at most 8 turns, windows of at most 60 rows sent as one header plus value arrays (half the tokens of JSON objects), at most 4 frames per VL call.
- **Measured:** the final run is 127 calls (18 to the VL model), 392k input and 38k output tokens, about $0.17. Each of the 3 repeat runs cost about the same ($0.17-0.18 by the meter, which over-prices the VL model on purpose).

15. What went wrong along the way

Problem	What happened	Change
Standstill flag	A rolling stop looked like a full stop	Stop details from wheel speed, in rules and validator
Skipped visual check	The model confirmed the red-leaves stop sign without looking, although the prompt asked it to	The rule moved from the prompt into the validator (vision gate)
No way to cite what VL saw	The model wanted to say "no stop sign" but could only cite OpenCV's reading	`frame_check` claims that must quote a real VL answer
Ambiguous VL answers	One "Yes" for four frames didn't say which frame showed what	One answer line per frame; a quote must come from that frame's line
Sideways frames	VL got raw rotated frames; lane recall 0.00	All VL frames undistorted and upright
Numbers in the reason	A reason said "~1.5 mph" where the data says 1.1; reasons aren't validated	Reasons carry no numbers; claims do
Unverified candidates	3 in run 1, mostly from an unclear "too many claims" error	The error says how many were given; 4 tries: 0 unverified
Labels vs definitions	My labels counted parking-lot moves my own definitions exclude	Fixed before the final scoring, with notes
MATLAB drift	The whole-drive replay fell 2.4 s behind	Stop waits retimed (section 13)

16. Limitations and next steps

- **One drive, 15 events.** The method and the evaluation are the contribution; the numbers are indicative.
- **Recall is bounded by the rules:** the agent only reviews what the rules propose; the hand-added event was never proposed.
- **Run-to-run variation** even at temperature 0. A majority vote over several runs is an obvious next step, to be chosen and measured on a new drive, not this one.
- **The model's judgement is not validated**, only its facts; the two failure modes of section 11 are the next fixes, to be measured on a second drive.
- **Scenarios hold only the ego car.** The vision already detects vehicles; adding them as actors is next.
- **Scale:** the batch steps are independent per drive and would run as a Nebius Serverless Job over a fleet.

17. Appendix: commands and files

```
# from the repository root, with the drive unzipped into test_sessions/
D=test_sessions/20261022T164127Z-084b
python src/scripts/agent.py $D --replay                          # the final run, from its recordings
(no key)
python src/scripts/score.py runs/20261022T164127Z-084b/agent/events.json
data/eval/events_gt_20261022T164127Z-084b.csv
python src/scripts/score_runs.py data/eval/events_gt_20261022T164127Z-084b.csv \
    runs/20261022T164127Z-084b/agent/events.json runs/20261022T164127Z-
084b/agent_repeats/run*/events.json
python src/scripts/export_matlab.py $D --labels data/eval/events_gt_20261022T164127Z-084b.csv
python src/scripts/verify_matlab.py                              # needs MATLAB + Automated Driving
Toolbox
python src/scripts/label_tool.py $D                              # watch the drive with the labels
python src/scripts/agent.py $D --agent-dir runs/mine --max-usd 0.5   # live, with NEBIUS_API_KEY
# recompute steps 1-3 (needs the vision weights): vision_batch.py, fuse.py, candidates.py
for t in src/tests/test_*.py; do python -m unittest "$t"; done
```

File	Role
`lld.py`, `signs.py`	the OpenCV pipeline from the OpenCV entry, unchanged
`src/scripts/vision_batch.py`, `fuse.py`, `candidates.py`	drive → vision → timeline → candidates
`src/scripts/agent.py`, `tokenfactory.py`	agent, tools, validator; Token Factory client with record, replay and cap
`src/scripts/score.py`, `score_runs.py`	scoring one run; spread over several
`src/scripts/label_events.py`, `label_tool.py`, `data/eval/`	labels, definitions, labelling tool
`src/scripts/export_matlab.py`, `verify_matlab.py`, `scenarios/`	MATLAB scenarios and their check
`src/scripts/eval_vl_lanes.py`	VL vs the 95 lane labels
`runs/20261022T164127Z-084b/`	every intermediate output; `agent/` = final run, `agent_repeats/` = noise runs